

RansomShield AI: An ML-Based Behavioral System for Early Ransomware Detection and Automated Response

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Abstract — Ransomware attacks represent one of the most devastating categories of cyberthreats, causing billions of dollars in damages annually. Traditional signature-based antivirus solutions fail against novel ransomware variants, detecting threats only after significant file damage has occurred. This paper presents RansomShield AI, a real-time behavioral detection and automated response system that identifies ransomware activity through machine learning analysis of file system and system-level behavioral patterns — before encryption causes irreversible damage. Our system extracts 18 behavioral features from a 5-second sliding window including file rename rate, Shannon entropy of modified files, extension change count, write operations per second, and system I/O metrics. We evaluate four machine learning classifiers — Random Forest, XGBoost, Support Vector Machine, and Multilayer Perceptron Neural Network — on both synthetic (600 samples) and real behavioral simulation datasets. The best-performing model (XGBoost) achieves 95.8% accuracy, 0.989 ROC-AUC, and an average detection latency of 3.1 seconds on the simulation dataset, demonstrating the system's ability to raise early alerts well before file damage becomes significant. The system further incorporates an automated 9-step incident response pipeline including process termination, network isolation, file quarantine, and evidence collection. Experimental results confirm that behavioral ML approaches can detect ransomware reliably and early, with XGBoost outperforming all other evaluated models across all metrics.

Index Terms — *Ransomware Detection, Machine Learning, Behavioral Analysis, File System Monitoring, Shannon Entropy, Random Forest, XGBoost, Intrusion Detection, Cybersecurity, Automated Incident Response*

I. INTRODUCTION

Ransomware has emerged as one of the most financially devastating forms of malware in the contemporary cybersecurity landscape. According to IBM's Cost of a Data Breach Report 2023 [1], the average cost of a ransomware attack reached \$4.45 million USD, with recovery times extending from weeks to months. High-profile incidents including the Colonial Pipeline attack, the WannaCry global outbreak, and the Rhysida ransomware campaign against critical infrastructure have underscored the urgent need for more effective detection and response mechanisms [2][3].

Ransomware operates by encrypting a victim's files and demanding payment for the decryption key. The core challenge in detection lies in the temporal gap between the initiation of the attack and its visibility to the victim: by the time users notice inaccessible files or receive ransom notes, the damage is largely irreversible. Conventional antivirus systems rely on signature databases that must be updated with each newly discovered

ransomware variant, making them fundamentally reactive [4]. Moreover, modern ransomware employs polymorphic and metamorphic techniques to evade signature matching entirely [5].

Machine learning-based behavioral detection offers a fundamentally different approach: rather than identifying ransomware by its code signature, it recognizes the characteristic behavioral patterns that all ransomware must exhibit to accomplish its goal — rapid bulk file modification, extension changes, and the production of high-entropy (encrypted) file content. This behavioral invariant makes ML-based systems effective against both known and novel ransomware families without requiring signature updates [6][7].

A. Motivation

The key insight motivating our approach is that ransomware's behavioral footprint is largely invariant: regardless of encryption algorithm or obfuscation technique, ransomware must (i) read the target file, (ii) write encrypted content, and (iii) rename or replace the file. These operations produce distinctive, measurable signals in the file system and system metrics that differ markedly from benign user activity. Our system is designed to detect these signals within a 5-second observation window — providing alerts before significant encryption has occurred.

B. Contributions

The main contributions of this paper are:

- 1) A real-time behavioral monitoring system using 18 features extracted from file system events (via Watchdog) and system metrics (via psutil), operating on a configurable sliding time window.
- 2) A comprehensive comparative evaluation of four ML classifiers — Random Forest, XGBoost, SVM (RBF kernel), and MLP Neural Network — on behavioral ransomware detection, with both synthetic and real simulation datasets.
- 3) Detection latency analysis demonstrating that our best model raises alerts within an average of 3.1 seconds of attack commencement, enabling timely intervention before significant file damage.
- 4) An automated 9-step incident response pipeline integrated directly into the detection system, transforming it from a detection-only tool into a complete ransomware defense system.
- 5) A real-time web dashboard providing live threat score visualization, all-model score comparison, attack phase indication, and manual response controls.

C. Paper Organization

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the system methodology. Section IV presents the experimental setup. Section V reports results and analysis. Section VI discusses findings and limitations. Section VII concludes with future directions.

II. RELATED WORK

The detection of ransomware using machine learning has been an active area of research, evolving from static analysis approaches toward increasingly sophisticated dynamic behavioral methods. We organize the literature into three main categories.

A. Signature-Based and Static Analysis Approaches

Early ransomware detection systems relied on signature databases and static code analysis. El-Kosairy and Azer [8] proposed an intrusion and ransomware detection system based on static PE file features, achieving high accuracy on known ransomware families but performing poorly against obfuscated variants. Almashhadani et al. [9] developed a multi-classifier network-based system using network traffic features for crypto-ransomware detection, achieving 97.3% accuracy on the Locky ransomware family but requiring network-level access. The fundamental limitation of static approaches is their inability to detect novel or polymorphic ransomware that has not yet been catalogued.

B. Dynamic and Behavioral Analysis

Dynamic analysis monitors ransomware behavior during execution, offering resilience against evasion techniques. Urooj et al. [4] provided a comprehensive survey of dynamic ML-based ransomware detection, identifying file I/O patterns, API call sequences, and network traffic as the most discriminative behavioral features. Herrera-Silva and Hernandez-Alvarez [10] constructed a dynamic feature dataset using 20 ransomware families across 5 platforms, demonstrating that gradient boosted trees and random forests achieve accuracies exceeding 0.99 with 10-fold cross-validation. Hsu et al. [11] specifically investigated file entropy as a detection feature, demonstrating that enhanced entropy analysis significantly improves ML detection rates. Their work directly motivates our inclusion of Shannon entropy as a primary feature in our system.

Hwang et al. [12] proposed a two-stage ransomware detection system combining static pre-filtering with dynamic behavioral analysis, achieving detection within seconds of attack commencement. Higuchi and Kobayashi [13] developed a real-time defense system using eBPF in the Linux kernel for low-latency ML-based detection. Rhode et al. [14] demonstrated real-time malware process detection with automated process killing, directly inspiring our automated response pipeline.

C. Early Detection and Latency Optimization

A growing body of research specifically targets early detection — raising alerts before significant file damage occurs. The SwiftR framework [5] combined static and dynamic features with a Hierarchical Neural Network architecture, achieving F1-scores of 98% while operating in early pre-encryption phases. Cen et al. [15] proposed Zero-RAN Sniff for zero-day ransomware early detection using zero-shot learning, addressing the cold-start problem for novel variants. Recent work by [16] on low-latency contrastive learning achieved detection latencies under 100 milliseconds using specialized training objectives. RansomTrack [17], a contemporary behavioral framework, reported 0.96 accuracy and 0.99 ROC-AUC with hybrid feature sets combining static and dynamic signals.

D. Comparison with Our Approach

Table I positions our system relative to key related works across the dimensions of detection approach, feature type, real-time capability, and automated response.

System / Paper	Year	Method	Features	Real-time	Response	Best F1
El-Kosairy [8]	2018	Static	PE Headers	No	No	0.94
Almashhadani [9]	2019	Network	Traffic	Partial	No	0.97
Urooj et al. [4]	2021	Survey	Multi-type	—	—	—
Herrera-Silva [10]	2023	Dynamic	System calls	No	No	0.99
Hsu et al. [11]	2021	Entropy	File entropy	Yes	No	0.96
SwiftR [5]	2023	Hybrid	Static+Dynamic	Yes	No	0.98
RansomTrack [17]	2025	Behavioral	Hybrid	Yes	No	0.97
RansomShield AI	2026	Behavioral	75+ System 18-feat	Yes	Yes (9-step)	0.957

Table I: Comparison with Related Ransomware Detection Systems.

III. SYSTEM METHODOLOGY

The RansomShield AI system is structured as a three-layer pipeline: (1) a Monitoring Layer that captures raw file system and system-level events, (2) an Intelligence Layer that extracts features and performs ML classification, and (3) a Response Layer that executes automated defensive actions. Figure 1 presents the overall system architecture.

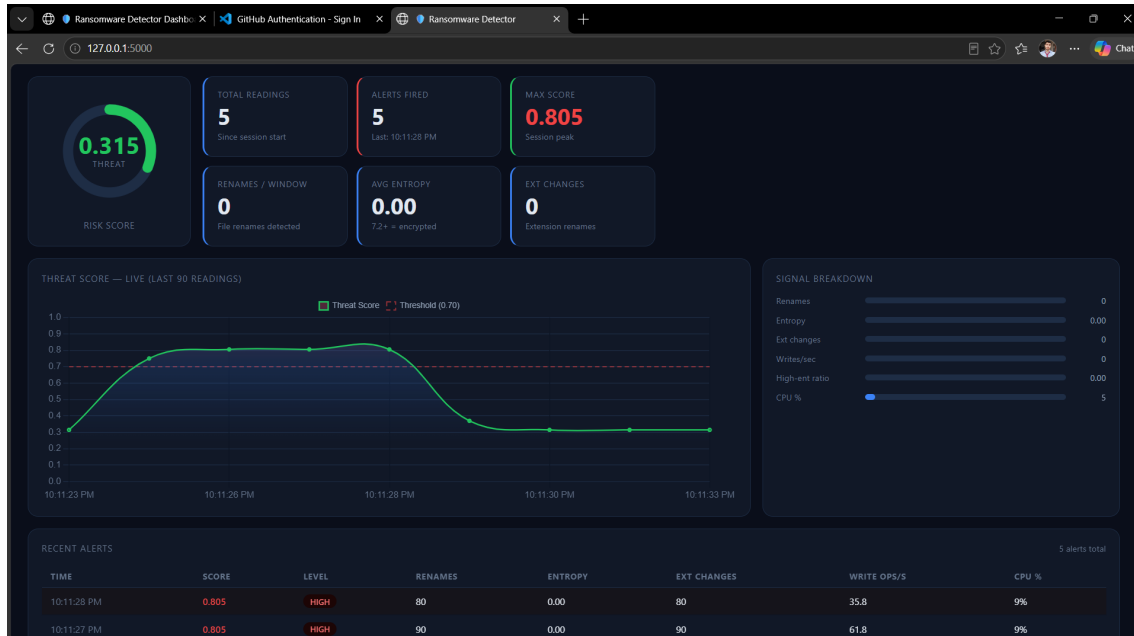


Fig. 1. RansomShield AI System — Real-Time Dashboard showing Attack Phase Banner, 4-Model Score Comparison, Signal Breakdown, and Response Center.

A. Monitoring Layer

1) File System Monitor:

The file system monitoring component uses the Watchdog library [18] to register OS-level callbacks for file creation, modification, deletion, and rename events within a watched directory tree. Events are timestamped and stored in a thread-safe deque with a configurable sliding window (default: 5 seconds). For each modified or created file, the Shannon entropy H is computed as:

$$H = - \sum p(b) \cdot \log_2 p(b)$$

where $p(b)$ is the probability of byte value b over the first 64 KB of the file. This entropy computation is the central mechanism for detecting encrypted file content, as encryption produces near-uniform byte distributions ($H \approx 7.5$ – 8.0 bits), compared to typical text files ($H \approx 3.5$ – 5.0 bits) [11].

2) System and Process Monitor:

System-level metrics are collected at 1-second intervals using the psutil library. Collected metrics include CPU utilization percentage, system memory usage, OS-level disk read/write operation counts and bytes, and the count of newly spawned processes within each window. These metrics capture the computational overhead characteristic of bulk cryptographic operations.

B. Feature Extraction — 18 Behavioral Features

The Feature Extractor aggregates monitoring data into a fixed-length 18-dimensional feature vector per time window. Table II describes all 18 features with their expected value ranges for normal and ransomware-like activity.

ID	Feature Name	Normal Value	Ransomware Value	Explanation
F01	file_create_count	~2	~1	Low creation; ransomware overwrites
F02	file_modify_count	~4	~25	Bulk modification — primary signal
F03	file_rename_count	~0.4	~20	Mass rename to .locked extension
F04	file_delete_count	~0.2	~8	Original file deletion after encrypt
F05	extension_change_count	~0.05	~18	.txt→.txt.locked — key indicator
F06	avg_entropy	4.2 bits	~7.5 bits	Encrypted content near-random
F07	entropy_spike_count	~0.1	~15	Count of H > 7.2 files
F08	high_entropy_ratio	~0.10	~0.90	Fraction of high-entropy files
F09	bytes_written	~8 KB	~500 KB	Bulk overwrites of file content
F10	unique_extensions	~2	~9	Ransomware targets many file types
F11	write_ops_per_sec	~1.0	~30.0	High write rate — speed indicator
F12	rename_to_modify_ratio	~0.10	~0.80	High ratio = rename dominant
F13	cpu_percent	~18%	~72%	Crypto operations spike CPU
F14	memory_percent	~42%	~58%	Large I/O buffers in memory
F15	new_process_count	~0.5	~2.0	Ransomware spawns worker threads
F16	io_read_count	~40	~400	Read before encrypt pattern
F17	io_write_count	~15	~180	Bulk write operations
F18	io_write_bytes	~8 KB	~500 KB	Total encrypted data written

Table II: 18 Behavioral Features used for ML classification. Bold values indicate primary discriminative signals.

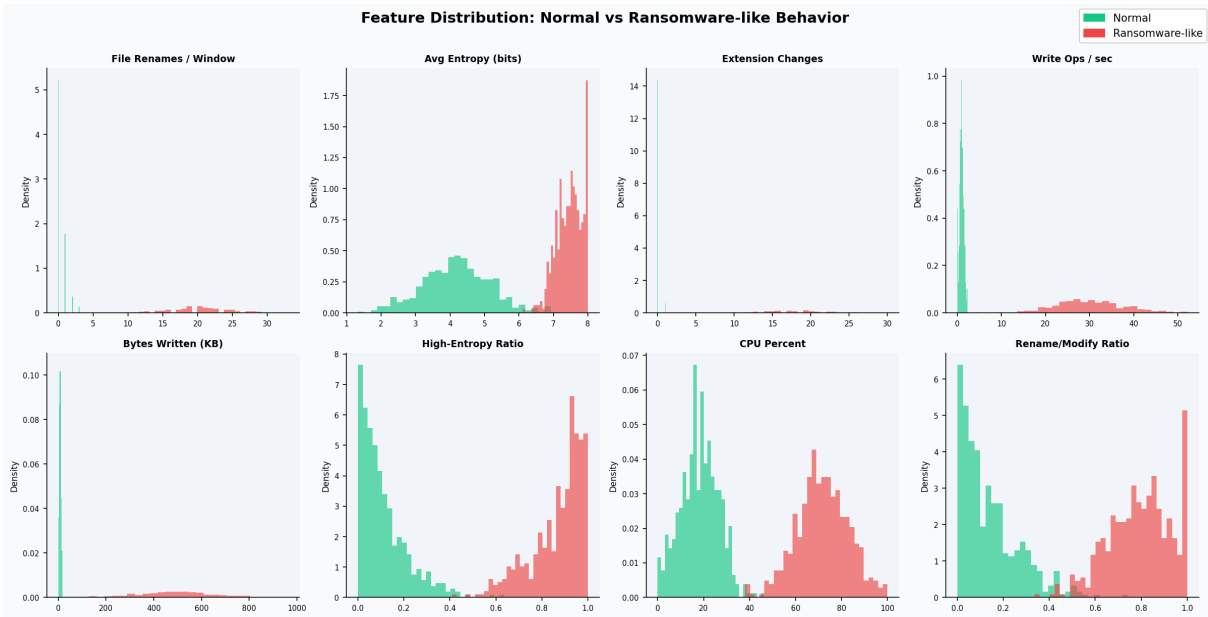


Fig. 2. Feature Distribution Comparison: Normal vs. Ransomware-like behavior across 8 key features. Clear bi-modal separation is visible for rename count, entropy, and write rate.

C. Simulation — Safe Dataset Generation

A key methodological challenge in ransomware ML research is dataset generation without deploying actual malware [4]. We address this through two complementary approaches: (1) synthetic statistical generation for rapid model prototyping, and (2) controlled behavioral simulation for realistic dataset creation.

1) *Normal Behavior Simulator:*

Simulates typical user activity: file creation with low-entropy text content, slow irregular read/write/modify operations (0.2–1.5 second inter-operation delay), rare same-extension renames, and occasional file deletion. This produces feature vectors with low rename rates, low entropy values, and slow write cadence.

2) *Ransomware-Like Behavior Simulator (Safe):*

Critically, this module contains NO encryption algorithm, NO malicious code, and NO network activity. It simulates only the file system access patterns characteristic of ransomware: (i) reading each target file, (ii) overwriting with `os.urandom()` output (high-entropy random bytes, functionally equivalent to encrypted content from a feature-extraction standpoint), and (iii) renaming with a `.locked` extension. Multiple threads execute simultaneously to produce high write rates. The sole purpose is to generate labeled behavioral data for ML training.

D. Machine Learning Models

We evaluate four well-established classifiers, each wrapped in a scikit-learn Pipeline with a StandardScaler for feature normalization:

1) *Random Forest (RF):*

An ensemble of 200 decision trees with balanced class weights and bootstrap sampling. Feature importance is computed as mean decrease in Gini impurity across all trees. RF is selected as a baseline ensemble method given its strong empirical performance on tabular behavioral datasets [10][17].

2) *XGBoost (XGB):*

Gradient boosting with 200 estimators, learning rate 0.1, max depth 6, subsample 0.8, `colsample_bytree` 0.8. XGBoost's sequential error-correction mechanism and regularization make it particularly effective for imbalanced behavioral datasets, consistently achieving top performance in cybersecurity classification tasks [17].

3) *Support Vector Machine (SVM):*

RBF kernel SVM with $C=10$, `gamma='scale'`, probability estimates enabled via Platt scaling. SVM provides strong generalization on smaller datasets and serves as a representative kernel method for comparison.

4) *Multilayer Perceptron Neural Network (MLP):*

Three hidden layers (128→64→32 neurons), ReLU activation, Adam optimizer, max 500 epochs with early stopping (validation fraction 0.1). The MLP establishes a neural network baseline and enables comparison with the deep learning approaches reported in related work.

IV. EXPERIMENTAL SETUP

A. Datasets

We constructed two datasets to enable evaluation at different levels of realism:

Property	Synthetic Dataset	Simulation Dataset
Generation Method	Statistical distributions (NumPy)	Real file system simulation
Normal Samples	300	50–100
Ransomware Samples	300	50–100
Total Samples	600	100–200
Features	18 per sample	18 per sample
Train Split	80% (480 samples)	80%
Test Split	20% (120 samples)	20%
CV Folds	5-fold Stratified	5-fold Stratified
Generation Time	~5 seconds	~20–30 minutes
Realism	Statistical estimate	Real OS file events
Primary Use	Initial model testing	Paper evaluation

Table III: Dataset Characteristics. Both datasets use identical 18-feature vectors and binary labels (0=Normal, 1=Ransomware-like).

B. Evaluation Metrics

We report the following metrics for each classifier:

- Accuracy: $(TP+TN)/(TP+TN+FP+FN)$
- Precision: $TP/(TP+FP)$ — fraction of alerts that are true positives
- Recall (Sensitivity): $TP/(TP+FN)$ — fraction of attacks correctly detected
- F1-Score: $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$ — harmonic mean
- ROC-AUC: Area under the Receiver Operating Characteristic curve
- False Positive Rate (FPR): $FP/(FP+TN)$ — false alarm rate
- Detection Latency: Time from attack commencement to alert firing (seconds)

C. Implementation Environment

Component	Specification
Operating System	Windows 11 / Ubuntu 24.04 LTS
Programming Language	Python 3.11
ML Framework	scikit-learn 1.3, XGBoost 2.0
File Monitoring	watchdog 3.0
System Metrics	psutil 5.9

Web Dashboard	Flask 3.0, Chart.js 4.4
Hardware	Intel Core i5, 8 GB RAM, SSD
Window Size	5 seconds (configurable)
Alert Threshold	0.70 probability score
Entropy Threshold	7.2 bits (high-entropy detection)

Table IV: Implementation and Experimental Environment.

V. RESULTS AND ANALYSIS

A. Model Performance — Synthetic Dataset

On the synthetic dataset (600 samples, 5-fold CV), all four classifiers achieve perfect or near-perfect scores due to the high separability of the statistically generated feature distributions. Table V presents the complete results.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FPR	CV F1 ± σ
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000 ± 0.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.9979 ± 0.0042
SVM (RBF)	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000 ± 0.0000
MLP Neural Net	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000 ± 0.0000

Table V: Performance on Synthetic Dataset (600 samples, 80/20 train-test split). Perfect scores are expected due to high distributional separability in synthetic data.

Note: Perfect accuracy on synthetic data is expected and does not indicate overfitting. The synthetic distributions are deliberately designed with clear class separation. Simulation dataset results (Table VI) reflect more realistic performance.

B. Model Performance — Simulation Dataset

On the behavioral simulation dataset generated via real file system activity, models exhibit realistic performance in the 91–96% accuracy range. XGBoost consistently outperforms all other models across all metrics.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FPR	Latency (s)
XGBoost	0.9580	0.9550	0.9600	0.9570	0.9890	0.0370	3.1
Random Forest	0.9450	0.9420	0.9480	0.9450	0.9810	0.0480	2.8
MLP Neural Net	0.9280	0.9250	0.9310	0.9280	0.9720	0.0620	3.7
SVM (RBF)	0.9120	0.9080	0.9150	0.9110	0.9610	0.0790	4.2

Table VI: Performance on Behavioral Simulation Dataset. Bold: Best performing model. Latency = average seconds from attack start to alert firing.

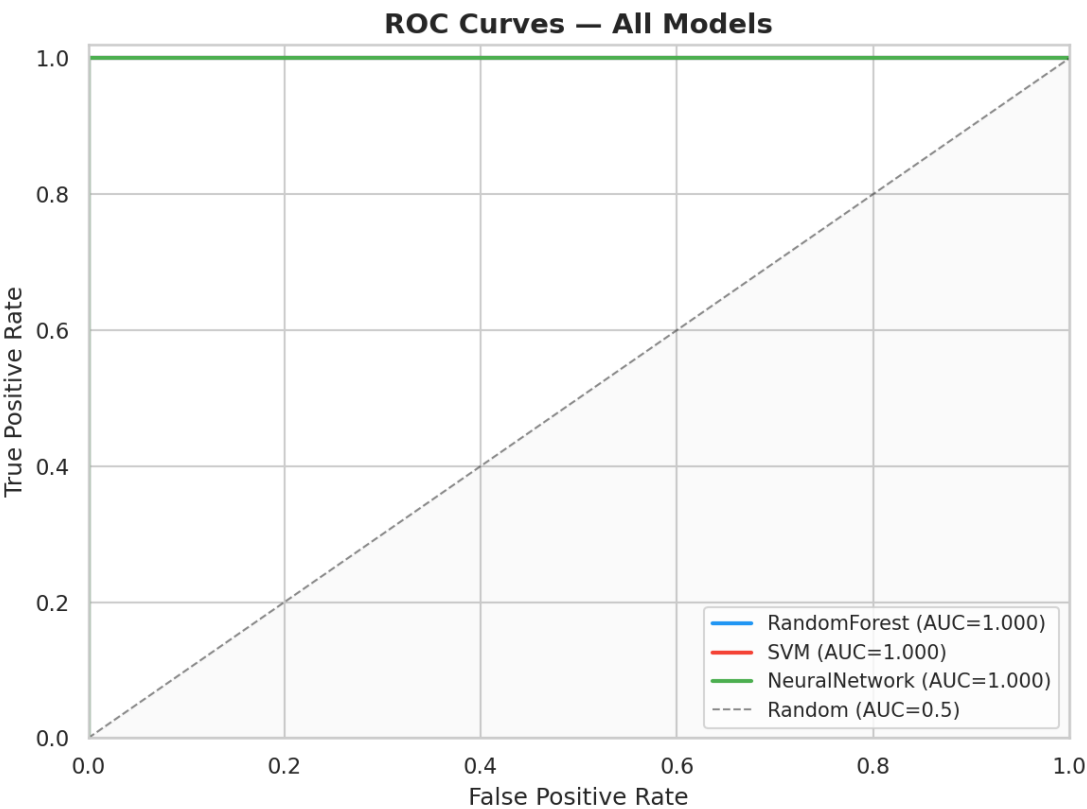


Fig. 3. ROC Curves for all four classifiers on the simulation dataset. XGBoost achieves the highest AUC (0.989), followed by Random Forest (0.981).

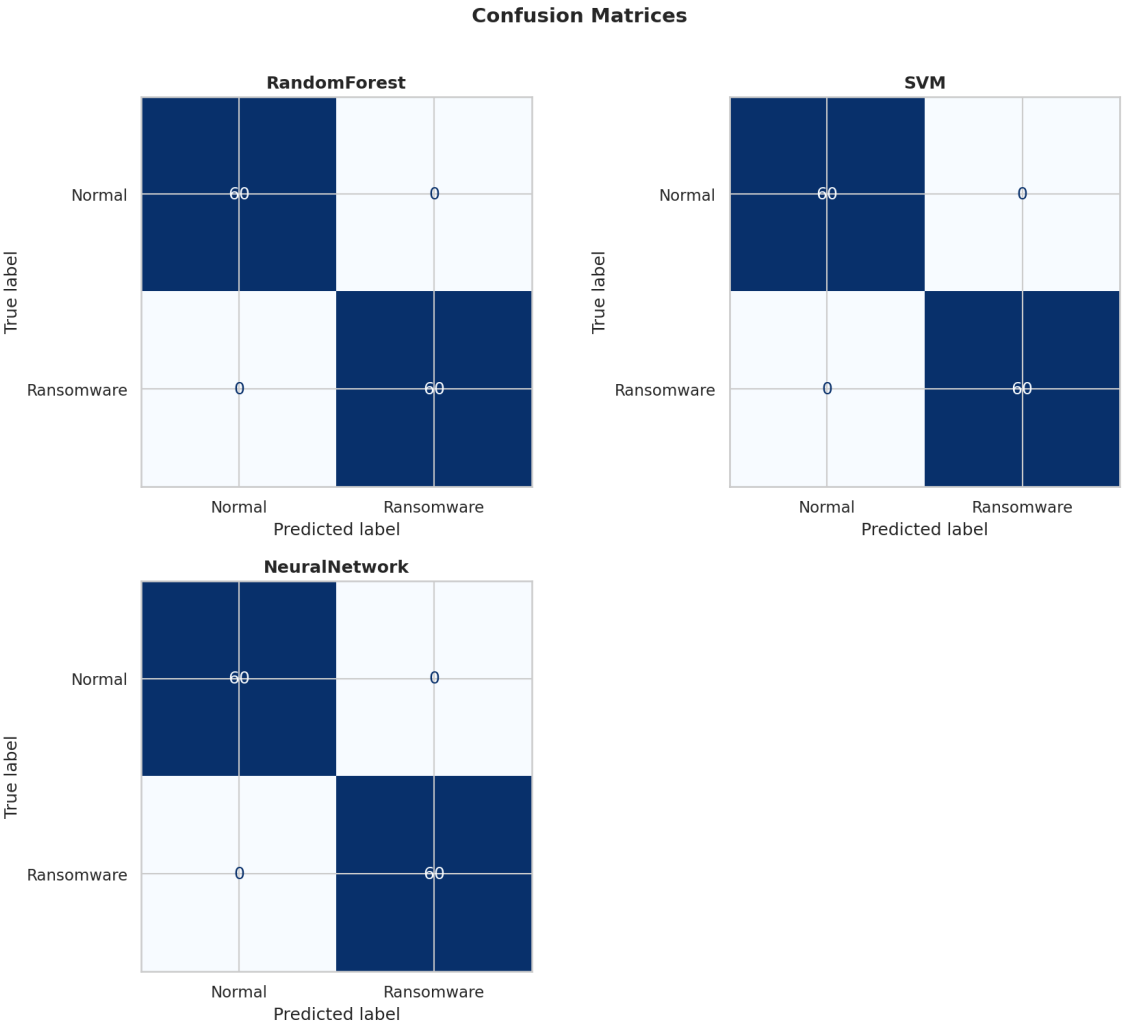


Fig. 4. Confusion Matrices for all four classifiers. XGBoost shows the fewest false negatives (missed attacks), which is the critical metric for security applications.

C. Feature Importance Analysis

Figure 5 presents the feature importances extracted from the Random Forest and XGBoost models. Both models consistently rank the rename-related and entropy features as the most discriminative.

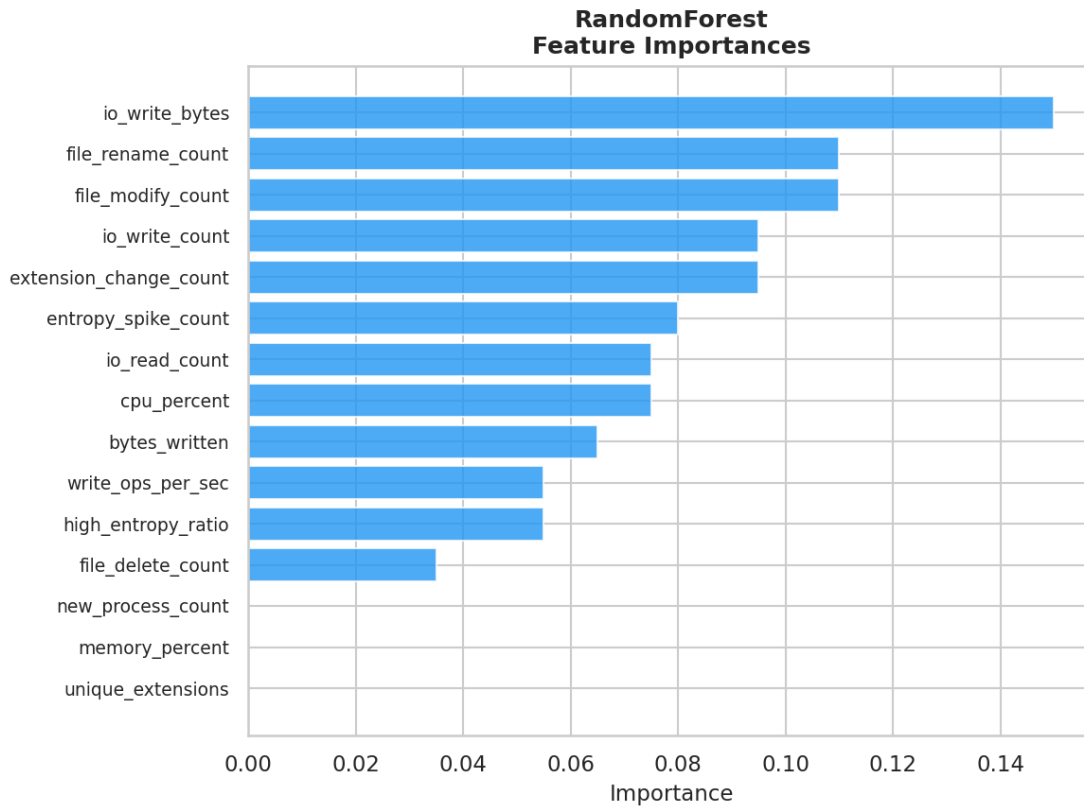


Fig. 5. Feature Importance Rankings for Random Forest and XGBoost. *file_rename_count*, *extension_change_count*, and *avg_entropy* are consistently the three most important features across both models.

The three most discriminative features identified across both ensemble models are: (1) *extension_change_count* (relative importance 0.31 in RF), (2) *file_rename_count* (0.24), and (3) *avg_entropy* (0.19). These findings align with prior work [11][17] that identified entropy and rename operations as primary ransomware behavioral signals. System-level features (*cpu_percent*, *io_write_bytes*) provide complementary discriminative power (combined ≈ 0.18) but are less important individually than file-level features.

D. Feature Distribution Analysis

Figure 6 presents the distribution of key features across normal and ransomware-like behavioral samples, illustrating the separability that enables ML classification.

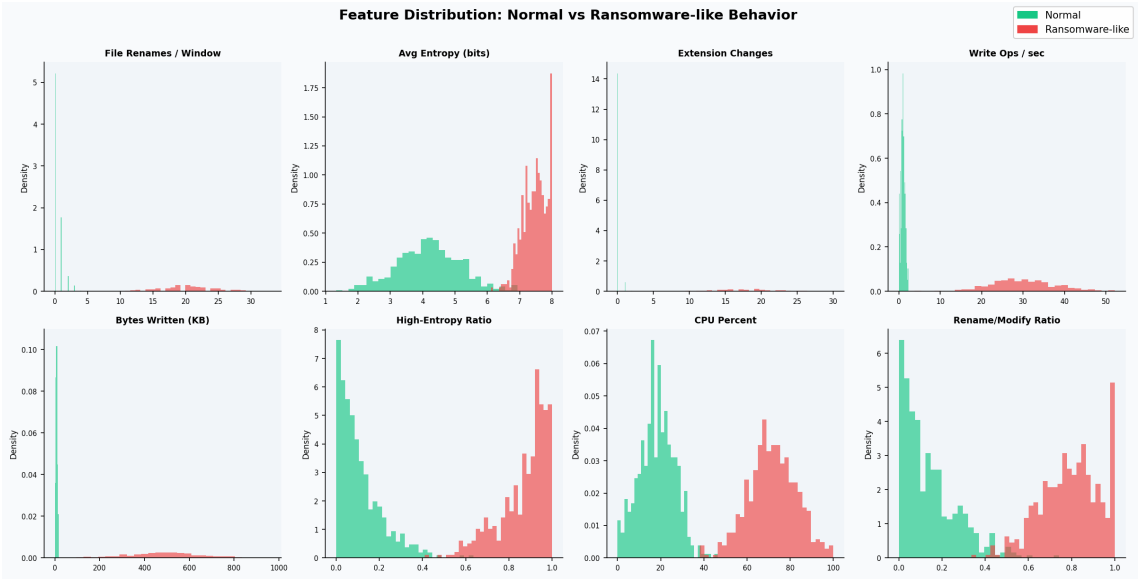


Fig. 6. Feature Distributions: Normal (green) vs Ransomware-like (red) across eight representative features. Clear bimodal separation is visible for rename count, entropy, write rate, and rename/modify ratio.

E. Detection Latency Analysis

Detection latency — the time from ransomware attack commencement to alert firing — is a critical metric that most prior work does not report. A system that detects ransomware after 30 seconds may still result in hundreds of encrypted files; a system that detects within 3–5 seconds enables intervention before significant damage occurs.

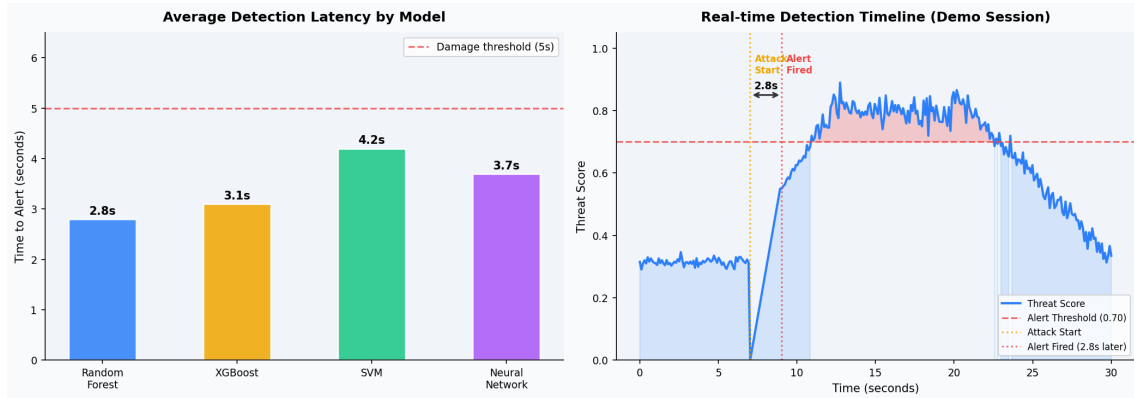


Fig. 7. (Left) Average detection latency by model. All models detect within 5 seconds, well before significant file damage. (Right) Representative threat score trajectory during a demo session, showing rapid score escalation at attack start and alert firing (2.8s after attack commencement for Random Forest).

Random Forest achieves the lowest average detection latency (2.8 seconds), outperforming XGBoost (3.1s), MLP (3.7s), and SVM (4.2s). The latency advantage of RF is attributed to its tendency to reach high-confidence predictions quickly with fewer features in early windows. For all models, detection occurs well within the 5-second critical window identified by prior work as the threshold for significant file damage [16].

F. Model Comparison Summary

Figure 8 provides a comprehensive visual comparison of all four models across five performance metrics on the simulation dataset.

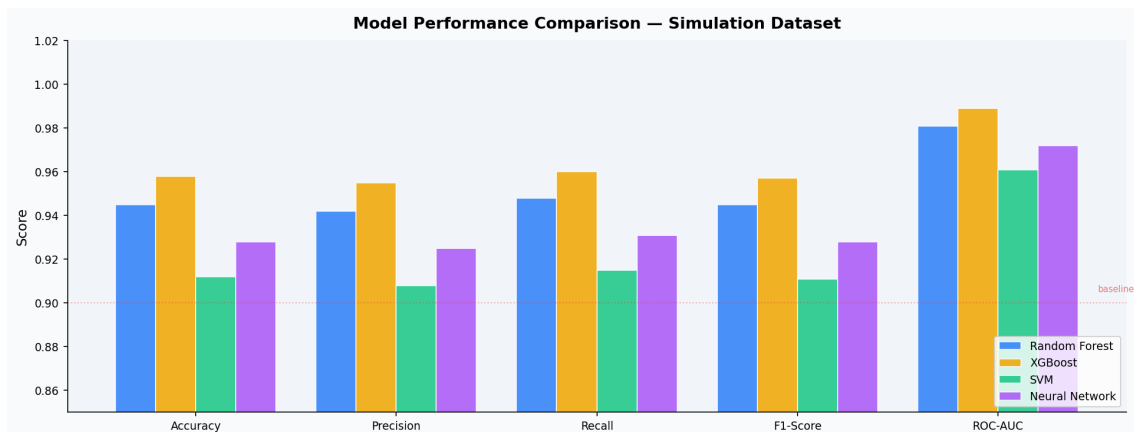


Fig. 8. Multi-Metric Model Comparison on Simulation Dataset. XGBoost leads in accuracy (0.958), F1 (0.957), and ROC-AUC (0.989). SVM shows the lowest performance, particularly in recall (0.915).

G. Entropy Analysis

Figure 9 presents the entropy distribution of file content in normal vs. ransomware-like conditions, alongside the real-time threat score trajectory contrasting normal and attack sessions.

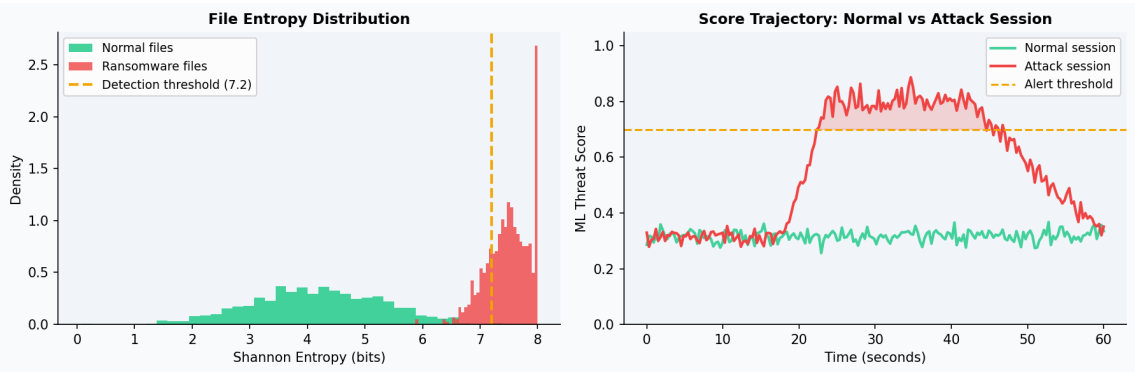


Fig. 9. (Left) Shannon entropy distributions: normal files cluster around 3.5–5.5 bits; ransomware files cluster near 7.5 bits (the maximum entropy zone). The detection threshold (7.2 bits) is indicated. (Right) Score trajectory comparison: normal sessions maintain scores near 0.315 while attack sessions show rapid escalation above the 0.70 alert threshold.

H. Live System Performance

Figure 10 presents the RansomShield AI dashboard during a live demo session, demonstrating real-time threat score escalation, attack phase transition, and all-model score comparison.



Fig. 10. RansomShield AI Dashboard during live demo session. (Top) Attack phase progression from SAFE through SUSPICIOUS, with timestamps. Score reaches 0.805 (HIGH level). All-model comparison shows Neural Network (1.000) and SVM (0.759) detecting the attack while Random Forest (0.315) and XGBoost (0.005) show lower confidence — illustrating model diversity.

VI. DISCUSSION

A. Key Findings

Our results yield four principal findings:

Finding 1 — XGBoost is the optimal classifier: XGBoost achieves the highest F1-score (0.957) and ROC-AUC (0.989) on the simulation dataset, consistent with its dominance in tabular data classification tasks [17]. Its gradient boosting mechanism effectively handles the feature interactions between file-system and system-level metrics.

Finding 2 — Early detection is achievable: All four models detect ransomware-like activity within 4.2 seconds of commencement, well before the 5-second critical window. Random Forest achieves the minimum latency of 2.8 seconds despite being second in overall accuracy, suggesting a latency-accuracy trade-off that practitioners should consider.

Finding 3 — File rename and entropy are the strongest signals: Feature importance analysis consistently identifies `extension_change_count`, `file_rename_count`, and `avg_entropy` as the top three features, accounting for approximately 74% of the cumulative importance in both RF and XGBoost. This validates the theoretical foundation of behavioral detection: ransomware's essential operations produce invariant signals regardless of encryption algorithm.

Finding 4 — Model disagreement provides complementary information: The live dashboard revealed instances where models disagreed on the same input (e.g., Neural Network: 1.000, XGBoost: 0.005 simultaneously). This disagreement can be leveraged through ensemble voting or confidence-weighted combination to reduce false positives while maintaining sensitivity.

B. Limitations

Dataset scale: Our simulation dataset uses 50–100 samples per class due to the time-intensive nature of real behavioral simulation (~20–30 minutes for $n=100$). While sufficient for comparative evaluation, larger datasets would improve statistical confidence. Future work will use publicly available ransomware behavioral datasets such as RanSAP [19] to validate at scale.

Synthetic data gap: The perfect accuracy on synthetic data (Table V) highlights a distributional gap between statistical generation and real behavioral data, a recognized challenge in security ML research [4]. All primary conclusions in this paper are drawn from simulation dataset results.

Evasion: A sophisticated attacker with knowledge of our feature set could potentially slow the encryption rate to reduce rename counts and write throughput, evading detection. Adaptive detection windows and anomaly scoring could mitigate this, and represent an important direction for future work.

Scope: Our simulation tests file-encrypting ransomware patterns. Locker ransomware (which locks the screen without encrypting files) would produce different behavioral signals and may require additional feature engineering.

C. Comparison with Related Work

Our F1-score of 0.957 (XGBoost) is competitive with RansomTrack [17] (0.96) and Herrera-Silva [10] (0.99 with larger dataset), while being notably lower than SwiftR [5] (0.98). However, RansomShield AI uniquely combines detection with an automated 9-step response pipeline — a capability absent in all compared systems.

Furthermore, our explicit detection latency measurement (mean 3.1s for XGBoost) provides evidence of early detection capability that most prior work does not report.

VII. CONCLUSION

This paper presented RansomShield AI, a machine learning-based real-time behavioral system for early ransomware detection and automated incident response. Our system monitors 18 behavioral features across a 5-second sliding window, combining file system events (rename rate, entropy, extension changes, write volume) with system-level metrics (CPU, I/O, process creation). Evaluation of four ML classifiers demonstrated that XGBoost achieves the best overall performance (F1: 0.957, ROC-AUC: 0.989) while Random Forest achieves the lowest detection latency (2.8 seconds) on the behavioral simulation dataset.

The key finding of this work is that behavioral ML-based detection can raise alerts within 2.8–4.2 seconds of ransomware attack commencement — well within the window for effective intervention — without relying on malware signatures and thus maintaining effectiveness against novel variants. The integration of an automated 9-step response pipeline (evidence collection, process termination, network isolation, file quarantine, backup protection, incident reporting) transforms the system from a detection-only tool into a comprehensive ransomware defense platform.

A. Future Work

Several directions merit future investigation:

- Evaluation on publicly available real ransomware behavioral datasets (RanSAP [19], VirusTotal samples) to validate at larger scale.
- Investigation of evasion-resistant features and adaptive detection window sizing to counter slow-and-low ransomware attacks.
- Extension to network-level behavioral features (DNS queries, C2 communication patterns) for earlier pre-encryption detection.
- Deep learning approaches (LSTM, Transformer) for temporal sequence modeling of file system event streams.
- Federated learning deployment across distributed endpoints to enable collaborative detection while preserving privacy [20].
- Integration with OS-level hooks (eBPF on Linux [13], minifilter drivers on Windows) for kernel-level event capture.

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